Course Roadmap

MSIS-2402 — Database Management Systems: Fundamentals of SQL

- Santa Clara University · Leavey School of Business
- 2 units · 5 weeks · 10 sessions of 95 minutes · Winter 2027
- Instructor: Dr. Mahmoud Parsian

🌍 Course Vision

Transform students from:

❌ spreadsheet users

➡️

✅ data thinkers who can query and explore data

🎯 Learning Outcomes

Students will be able to:

- Explain how relational data is organized (tables, keys, relationships)
- Create, retrieve, filter, summarize, and modify data with SQL
- Combine related tables and break complex questions into smaller queries
- Use SQL for data analysis, including window functions and CTEs
- Describe how databases fit into web and other modern applications
- Recognize the main types of modern data systems and the operational, security, performance, and governance issues in managing data

🧠 Learning Progression

```
Data

→ Query & Summarize          (Week 1)

→ Combine & Decompose        (Week 2)

→ Analyze                    (Week 3)

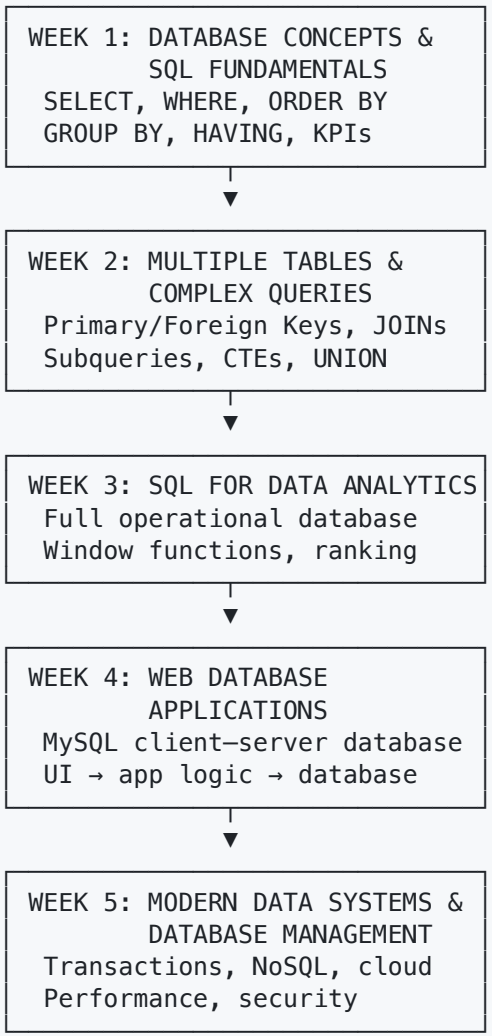
→ Build Applications         (Week 4)

→ Manage Data at Scale       (Week 5)
```

📊 Course Structure — 5 Weeks

Week	Theme	Skill
1	Database Concepts and SQL Fundamentals	Querying & summarizing
2	Working with Multiple Tables and Complex Queries	Combining & decomposing
3	SQL for Data Analytics	Insight discovery
4	Introduction to Web Database Applications	Application architecture
5	Modern Data Systems and Database Management	The data landscape

MSIS-2402 — Course Roadmap



Slide — Overall Course Progression

- Week 1 → Database Concepts and SQL Fundamentals
- Week 2 → Working with Multiple Tables and Complex Queries
- Week 3 → SQL for Data Analytics
- Week 4 → Introduction to Web Database Applications
- Week 5 → Modern Data Systems and Database Management

Goal of the course:

Learn how to **analyze real business data using SQL**, and understand how databases power applications and organizations.

🟦 Week 1 — Database Concepts and SQL Fundamentals

How relational data is organized, and the core skills for creating, retrieving, filtering, summarizing, and modifying data.

Topics

- What is a database / a relational database / an RDBMS
- What is SQL
- Tables, rows, columns

🟦 Week 1 — Retrieving & Filtering Data

Students learn:

- SELECT … FROM
- WHERE, AND / OR / NOT (`AND` runs before `OR` — use parentheses)
- ORDER BY
- LIMIT
- DISTINCT

Dataset:

A single **sales** table — 200 rows,
`week1_db_concepts_sql_fundamentals/part1_single_table_sql/data/sales.csv` .

SQL for Business Analytics

Data

→

1. Transform Data

→

2. Insight

→

3. Decisions

Week 1 — Example Query

The 10 largest sales (ties broken by `sale_id`).

```sql
SELECT *
FROM sales
ORDER BY total_amount DESC, sale_id
LIMIT 10;
```

Learning outcome:

Students learn to **retrieve and filter data**.

SQL Mindset

> SQL is NOT code

> SQL is a **question language**

🟦 Week 1 — Summarizing Data

- `COUNT`, `SUM`, `AVG`, `MIN`, `MAX` · `GROUP BY`, `HAVING`
- basic business metrics (KPIs)
- modifying data: `INSERT`, `UPDATE`, `DELETE`
- basic text and date functions

```
SELECT product_category,
       ROUND(SUM(total_amount), 2) AS total_sales
FROM sales
GROUP BY product_category
ORDER BY total_sales DESC;
```

Business Insight

- Which product category leads?
- Where to invest?
- What drives growth?

Week 2 — Multiple Tables and Complex Queries

Students move from **one table → multiple tables**.

Tables introduced:

- `customers`
- `orders`
- `products`
- `order_items`

Week 2 — Multiple Tables and Complex Queries (cont.)

Topics

- primary keys and foreign keys
- relational modeling and normalization
- INNER JOIN, LEFT JOIN, FULL OUTER JOIN
- multi-table joins (and why a wrong `ON` condition silently mismatches rows)
- subqueries, CTEs (`WITH`), `UNION` / `INTERSECT` / `EXCEPT`
- reasoning carefully about query results

Week 2 — Example Query

Orders by country.

```
SELECT c.country,
       COUNT(o.order_id) AS num_of_orders
FROM customers c
JOIN orders o
  ON c.customer_id = o.customer_id
GROUP BY c.country
ORDER BY num_of_orders DESC;
```

Learning outcome:

Students learn to **combine tables to answer questions**.

Week 2 — Example Query

Revenue by product category.

```
SELECT p.category,
       ROUND(SUM(oi.sales_amount), 2) AS total_revenue
FROM order_items oi
JOIN products p
  ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY total_revenue DESC;
```

Learning outcome:

Students perform their **first multi-table business analytics queries**.

Week 3 — SQL for Data Analytics

Students explore a **real business schema**.

Tables

```
customers
orders
order_items
products
stores
payments
shipments
```

Week 3 — SQL for Data Analytics (cont.)

Topics

- operational analytics and data quality checks
- window functions; ranking with `ROW_NUMBER`, `RANK`, `DENSE_RANK`
- Common Table Expressions (CTE)
- running totals, `LAG`, and trends

Week 3 — Example Query

Revenue by store type.

```
SELECT s.store_type,
       ROUND(SUM(oi.sales_amount), 2) AS total_revenue
FROM stores s
JOIN orders o ON s.store_id = o.store_id
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY s.store_type
ORDER BY total_revenue DESC;
```

Learning outcome:

Students analyze a **complete business database**.

Week 3 — Example Query

The 10 top customers by revenue.

```
SELECT o.customer_id,
       ROUND(SUM(oi.sales_amount), 2) AS revenue,
       RANK() OVER (
          ORDER BY SUM(oi.sales_amount) DESC
       ) AS rnk
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY o.customer_id
ORDER BY rnk, o.customer_id
LIMIT 10;
```

Learning outcome:

Students perform **real analytics workflows in SQL**.

Week 4 — Introduction to Web Database Applications

The role of databases in web and other modern applications.

Topics

- Client–server databases: MySQL, constraints, views (Session 7)
- Application architecture: user interface → application logic → database
- Basic web technologies: URLs, HTTP requests, forms, `GET` vs. `POST`
- A simple database-backed web application in Python + DuckDB (Session 8)

Week 5 — Modern Data Systems and Database Management

Beyond SQL and relational databases.

Topics

- Transactions, ACID, and concurrency — two live MySQL sessions (Session 9)
- Types of data systems: OLTP vs. OLAP, NoSQL, warehouses, lakes, cloud
- Performance: query plans (`EXPLAIN`) and indexes (Session 10)
- Security (`GRANT` , least privilege), governance, and backups

What You Will Be Able To Do

By the end of the course students can:

- explore unfamiliar databases
- write complex SQL queries
- join multiple tables
- perform data analytics using SQL
- answer real business questions
- explain how applications and organizations use and manage databases

SQL becomes a **data analysis tool**, not just a query language.

🧪 Tools & Technologies

- SQL
- Python
- **DuckDB** — the main SQL engine (Sessions 1–6 and 8, Lab 9A)
- **MySQL** — a client–server database (Sessions 7, 9, and 10)
- **Marimo** — reactive Python notebooks with SQL cells
- **qStudio** — a desktop SQL editor
- CSV datasets

🗓️ How Each Session Works

Time	What happens
First ~55 minutes	Lecture and live demo
Last ~40 minutes	**Lab A** in class — submit to Camino **"Session N Lab A"** before class ends
By **11:59 pm Pacific Time the day after class**	Submit **Lab B** to Camino **"Session N Lab B"**

Every session has a **Lab A** (core, 40 points, 6 tasks, done in class) and a **Lab B** (a step more challenging, 30 points, 6 tasks; −3 points per started 24 hours late) — published in the course repository's `LABS/` folder during class.

📦 Assessments

Component	Points
🧪 10 Labs — one per session, 70 points each (a 40 + b 30)	700
✏️ Final Exam — 2 hours, closed book and notes, lock-down browser, after Session 10	300
Total	**1,000**

Your lowest lab score is replaced by the average of the other nine.

Details: `course_information/ASSIGNMENTS_and_GRADING.md`

💼 Teaching Philosophy

- Learn by doing
- Focus on intuition over theory
- Emphasize business questions

🎯 Final Outcome

Students will:

- confidently query data
- understand datasets
- be ready for **MSIS-2407** — DBMS - Design, Development & Administration

🚀 Career Relevance

Prepares students for:

- Data Analyst
- Business Analyst
- Analytics roles

🙌 End of Roadmap

Let’s Begin

Week 1 starts with a simple question:

What can we learn from sales data?

Let’s start querying.